

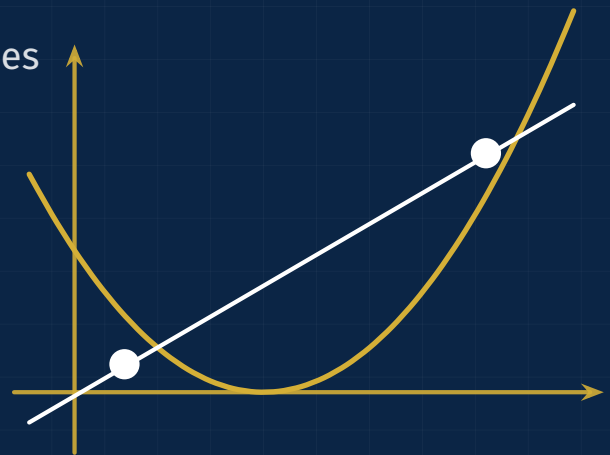


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CHAPTER 3 • SEQUENCES, FUNCTIONS & GRAPHS • TOPIC 9

Transformations of Graphs

Inside the bracket changes x and does the opposite; outside changes y and does what it says. Follow one point and you have the whole curve.



5

VARIANTS

50

PAST-PAPER PARTS

31

PRACTICE QUESTIONS

YOU WILL MASTER

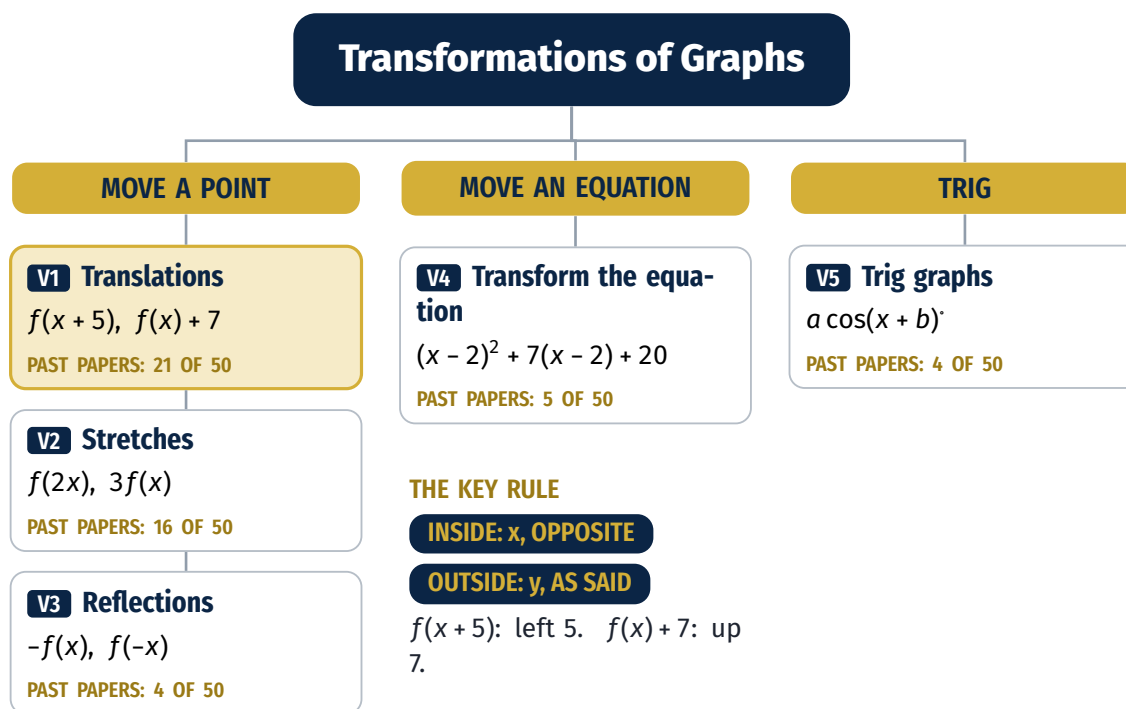
V1 Translations **V2** Stretches **V3** Reflections

V4 Transform the equation **V5** Trig graphs a, b, c

TOPIC MAP • EVERY VARIANT ON ONE PAGE

Transformations at a glance



Your notes


★ **How often does each variant come up?**

19 past-paper questions (Jan 2020 → Nov 2025) hold 50 parts; 31 practice questions, each counted under its main type.

	PAST PAPERS (OUT OF 50)	PRACTICE PARTS
V1 Translations	21	13
V2 Stretches	16	2
V3 Reflections	4	4
V4 Transform the equation	5	3
V5 Trig graphs	4	9

WHAT THIS TELLS YOU On almost every paper (Q19–25): “the turning point is $(5, -4)$ — write down the turning point of $f(x + 7)$, $f(x) - 6$, $f(2x)$ ”. One mark each, 30 seconds each — if you know the rule.

BEFORE YOU START

The big idea



Follow one point.

Every transformation moves each point the same way. Track the **turning point** (x, y) :

Change **inside** the bracket $\rightarrow$ the x changes, the **opposite** way.

Change **outside** $\rightarrow$ the y changes, exactly as written.

equation point (x, y) goes to

$$f(x + a) \quad (x - a, y) \text{ left } a$$

$$f(x) + a \quad (x, y + a) \text{ up } a$$

$$f(ax) \quad \left(\frac{x}{a}, y\right)$$

$$af(x) \quad (x, ay)$$

$$-f(x) \quad (x, -y)$$

$$f(-x) \quad (-x, y)$$



The four moves

Translate

$$\begin{pmatrix} -a \\ 0 \end{pmatrix} \text{ or } \begin{pmatrix} 0 \\ a \end{pmatrix}$$

Slide; same shape.

Stretch

$$\times a \text{ or } \div a$$

Pulled from an axis.

Reflect

$$-f(x), f(-x)$$

Mirror in an axis.

Combine

$$2f(x - 1)$$

One step at a time.

Same point, four results — turning point $(6, 5)$

$$f(x - 4)$$

$$(10, 5)$$

Right 4 (opposite of $-$).

$$f(x) - 4$$

$$(6, 1)$$

Down 4.

$$f(3x)$$

$$(2, 5)$$

$x \div 3$.

$$3f(x)$$

$$(6, 15)$$

$y \times 3$.



CALCULATOR FIRST, THEN CHECK

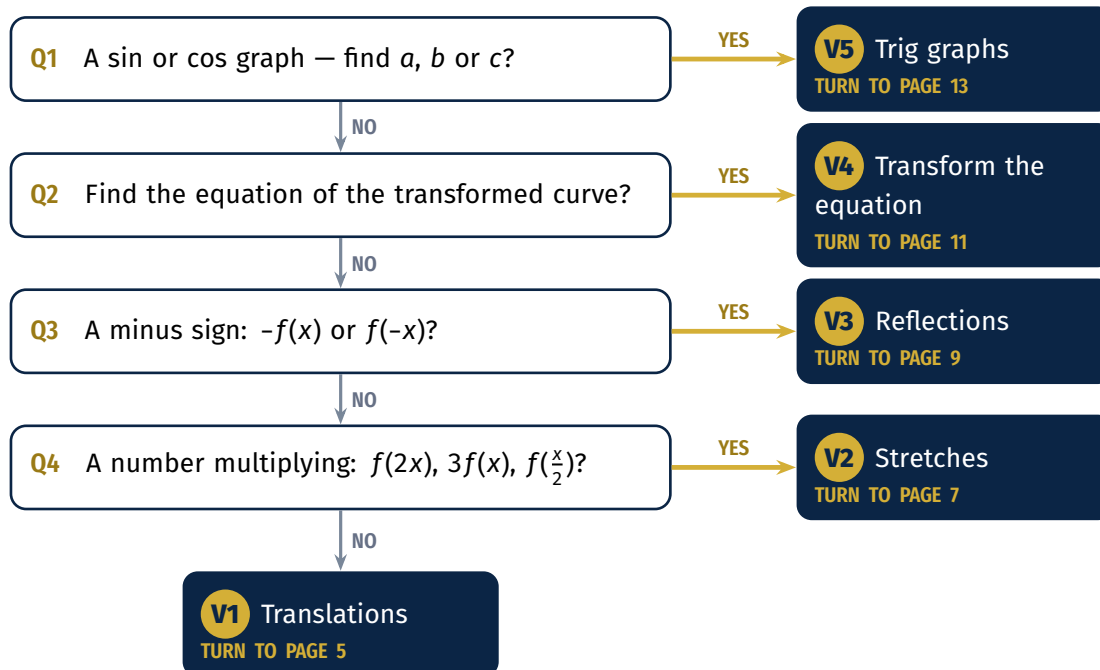
Check one point: if $(6, 5)$ is on $y = f(x)$, then on $y = f(3x)$ the point $(2, 5)$ works because $f(3 \times 2) = f(6) = 5 \checkmark$.

COUNT / RECOGNISE, THEN CHOOSE

Which variant am I?

 Your notes

Read the question, then walk down the ladder. The first **YES** tells you the variant — and the page to turn to.



COLOUR CODE IN THE WORKING

Blue = letter terms $4x$, $-3y$

Orange = plain numbers 17 , -13

STEP 1 **STEP 2** **STEP 3** **STEP 4** = which step of “How do I do it?” each line is.

ON EVERY VARIANT SPREAD



What is it?



When you see → you do



How do I do it?



Watch out



Worked example



Exam tip



Check yourself



Calculator check

V1**VARIANT 1** MOST COMMON IN THE EXAM

Translations

 $f(x + a)$: left a . $f(x) + a$: up a . Inside does the opposite.

SEEN IN PAST PAPERS

21 / 50

PRACTICE QUESTIONS: 13



Your notes

Came up in: Jan 2020 P1HR Q22 • Jan 2020 P2H Q21 • Jan 2021 P1H Q21 • Jan 2022 P1HR Q20 • Jan 2022 P1H Q23 • Jun 2022 P1H Q20 • Jun 2022 P2HR Q22 • May 2023 P1HR Q20 • May 2023 P1H Q21 • May 2024 P1H Q21 • Jun 2024 P2HR Q23 • Nov 2024 P2H Q19 • Jun 2025 P1HR Q21 • Jun 2025 P1H Q19 • Nov 2025 P1H Q24

? What is it?

In plain words

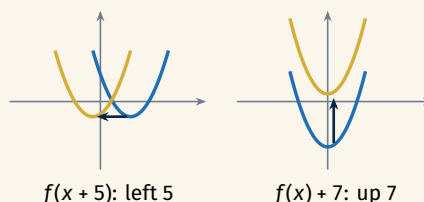
A translation **slides** the curve. $f(x) + 7$: every point goes **up 7**. $f(x + 5)$: every point goes **left 5** — inside the bracket does the **opposite** of the sign. Vector: $\begin{pmatrix} -5 \\ 0 \end{pmatrix}$.

IT LOOKS LIKE

$f(x + 5)$

$f(x) - 6$

$g(x + a) + b$



How do I do it?

1 Inside or outside?

Is the number in the bracket with x , or added after?

2 Inside

 x changes, opposite sign: $f(x + 5) \rightarrow x - 5$.

3 Outside

 y changes as written: $+7 \rightarrow y + 7$.

4 Write

The new point.

INSIDE: OPPOSITE $f(x - 2)$ moves right 2.

V1 Translations

VARIANT 1 · SEE IT DONE, THEN TRY


 Your notes
WHEN YOU SEE → YOU DO

"Write down the coordinates of the turning point of $y = f(x + 5)$ " → $x - 5$; y **unchanged**

"... of $y = f(x) - 6$ " → $y - 6$; x **unchanged**

" $y = g(x + a) + b$ maps $(4, -5)$ to $(1, -16)$ " → $4 - a = 1$, $-5 + b = -16$

WORKED EXAMPLE **PAST PAPER**

November 2025 · Paper 1H · Q24(i)(ii) · 2 marks

The only turning point of $y = f(x)$ is $(4, 3)$. Write down the turning point on the curve with equation (i) $y = f(x + 5)$ (ii) $y = f(x) + 7$.

STEP 1 $f(x + 5)$: inside ◀ x changes, opposite

STEP 2 $(4 - 5, 3) = (-1, 3)$ ◀ (i) left 5

STEP 3 $f(x) + 7$: outside ◀ y changes, as written

STEP 4 $(4, 3 + 7) = (4, 10)$ ◀ (ii) up 7

ANSWER (i) $(-1, 3)$ (ii) $(4, 10)$

✓ $f(-1 + 5) = f(4) = 3$ ✓

WATCH OUT

✗ $f(x + 5)$: $(9, 3)$

✓ $(-1, 3)$

Inside the bracket moves the opposite way.

EXAM TIP

Check: put the new x into the bracket — you must get the old x .

Check yourself

calculator first · answers on page 15

1.1 May 2024 P1H

Minimum $(5, -4)$. Find the minimum of (i) $y = f(x + 7)$ (ii) $y = f(x) - 6$.

Answer:

1.2 Jan 2022 P1H

Maximum $(5, 7)$. Find the maximum of (i) $y = f(x + 9)$ (ii) $y = f(x) + 3$.

Answer:

1.3 Jun 2025 P1HR

$y = g(x)$ becomes $y = g(x + a) + b$. The point $(4, -5)$ maps to $(1, -16)$. Find a and b .

Answer:

V2

VARIANT 2

Stretches

$f(ax)$: $x \div a$. $af(x)$: $y \times a$. Inside still does the opposite.

SEEN IN PAST PAPERS

16 / 50

PRACTICE QUESTIONS: 2



Your notes

Came up in: Jan 2020 P1HR Q22(b) • Oct/Nov 2020 P1HR Q20 • Oct/Nov 2020 P1H Q25(b) • Jan 2020 P2H Q21 • Jan 2021 P1H Q21 • Jan 2022 P1HR Q20 • Jun 2022 P1H Q20 • Jun 2022 P2HR Q22 • May 2023 P1HR Q20 • May 2023 P1H Q21 • Jun 2024 P2HR Q23 • Nov 2024 P2H Q19 • Jun 2025 P1HR Q21 • Jun 2025 P1H Q19 • Nov 2025 P1H Q24

? What is it?

In plain words

$3f(x)$: every y is **multiplied by 3** — taller.

$f(3x)$: every x is **divided by 3** — squashed towards the y -axis (inside: opposite, so $\times 3$ becomes $\div 3$). $f(\frac{x}{2})$: $x \times 2$.

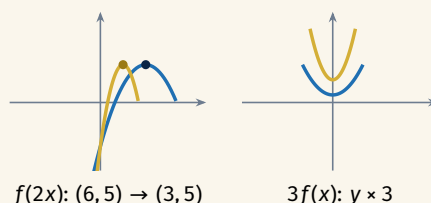
IT LOOKS LIKE

$$f(2x)$$

$$3f(x)$$

$$f(\frac{x}{2})$$

$$\frac{1}{3}f(x)$$



How do I do it?

1 Find the number

Inside ($f(3x)$) or outside ($3f(x)$)?

2 Inside

$x \div$ that number (opposite of $\times$).

3 Outside

$y \times$ that number.

4 Write

The other coordinate stays.

INSIDE $\div$, OUTSIDE $\times$ $f(\frac{x}{2})$: $x \times 2$.

V2 Stretches

VARIANT 2 · SEE IT DONE, THEN TRY

 Your notes
WHEN YOU SEE → YOU DO

"turning point of $y = f(3x)$ " → $x \div 3$, y **stays**

"... of $y = 3f(x)$ " or " $\frac{1}{3}f(x)$ " → $y \times 3$ **or** $y \div 3$, x **stays**

"image of A on $y = f(\frac{x}{2})$ " → $x \times 2$

**WORKED EXAMPLE** **PAST PAPER**

June 2022 · Paper 1H · Q20 · 2 marks

The maximum point of $y = f(x)$ is (s, t) . Find, in terms of s and t , the maximum point of (i) $y = f(x - 2)$ (ii) $y = 3f(x)$.

STEP 1 $f(x - 2)$: inside ◀ x changes, opposite

STEP 2 $(s + 2, t)$ ◀ (i) right 2

STEP 3 $3f(x)$: outside ◀ y changes, as written

STEP 4 $(s, 3t)$ ◀ (ii) $y \times 3$

ANSWER (i) $(s + 2, t)$ (ii) $(s, 3t)$

✓ same rule with letters ✓

**WATCH OUT**

✗ $f(2x)$: $(2x, y)$

✓ $(\frac{x}{2}, y)$

Inside does the opposite: $\times 2$ becomes $\div 2$.

**EXAM TIP**

$f(2x)$ and $2f(x)$ look alike — circle whether the 2 is inside or outside first.

**Check yourself**

calculator first · answers on page 15

2.1

Oct/Nov 2020 P1HR

Maximum $(-3, 4)$. Find the maximum of $y = f(2x)$.

Answer:

2.2

Jun 2022 P2HR

$A(-3, 2)$ is on $y = f(x)$. Find its image on $y = f(\frac{x}{2})$.

Answer:

2.3

Jan 2021 P1H

Minimum $(-9, 15)$. Find the minimum of $y = \frac{1}{3}f(x)$.

Answer:

V3

VARIANT 3

Reflections

$-f(x)$: flip in the x -axis ($y \rightarrow -y$). $f(-x)$: flip in the y -axis ($x \rightarrow -x$).

SEEN IN PAST PAPERS

4 / 50

PRACTICE QUESTIONS: 4

 Your notes

Came up in: Oct/Nov 2020 P1H Q25(a) • Nov 2021 P2H Q21(c) • Jun 2022 P2HR Q22(b) • May 2023 P1HR Q20(a)(iii)

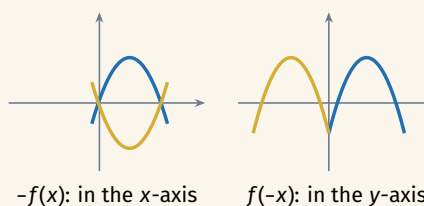
? What is it?**In plain words**

Minus **outside**: $-f(x)$ — every y changes sign:

reflection in the x -axis.

Minus **inside**: $f(-x)$ — every x changes sign:

reflection in the y -axis.

IT LOOKS LIKE $-g(x)$ $f(-x)$ $-g(x - c)$ **How do I do it?**

1 Where is the minus?

Outside: $-f(x)$. Inside: $f(-x)$.

2 Outside

$(x, y) \rightarrow (x, -y)$: in the x -axis.

3 Inside

$(x, y) \rightarrow (-x, y)$: in the y -axis.

4 Describe

“Reflection in the ... axis” — name the axis.

MINUS OUTSIDE: x -AXIS Minus inside: y -axis.

V3 Reflections

VARIANT 3 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

“Describe fully the transformation”
from $g(x)$ to $-g(x)$

→ **Reflection in the x-axis**

“... on the curve $y = f(-x)$ ”

→ $x \rightarrow -x$

“image of $B(p, q)$ on $y = -g(x - c)$ ”

→ **Right c , then flip:** $(p + c, -q)$ **WORKED EXAMPLE****PAST PAPER**

June 2022 · Paper 2HR · Q22(b) · 2 marks

The point $B(p, q)$ lies on $y = g(x)$. Find the coordinates of the image of B on the curve $y = -g(x - c)$, where c is a constant.

STEP 1 $g(x - c)$: inside ◀ right c

STEP 2 $(p + c, q)$ ◀ translate first

STEP 3 $-g(\dots)$: outside minus ◀ reflect in the x-axis

STEP 4 $(p + c, -q)$ ◀ flip the y

ANSWER $(p + c, -q)$ ✓ $-g(p + c - c) = -g(p) = -q$ ✓**WATCH OUT**

× $-f(x)$: “reflection in the y-axis”
✓ in the x-axis

Outside minus flips y, i.e. top to bottom.

**EXAM TIP**

“Describe fully” for a reflection needs the word reflection and the axis.

**Check yourself**

calculator first · answers on page 15

3.1

May 2023 P1HR

Maximum $(3, 5)$. Find the maximum of $y = f(-x)$.

Answer:

3.2

Nov 2021 P2H

C: $y = 9 - 3(x + 2)^2$ maps to **T:** $y = 3(x + 2)^2 - 9$. Describe the transformation.

Answer:

3.3

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Minimum $(2, -3)$. Find the minimum of $y = f(-x)$.

Answer:

V4

VARIANT 4

Transform the Equation

Translation $\begin{pmatrix} a \\ b \end{pmatrix}$: replace x by $(x - a)$ and add b — then expand if asked.

SEEN IN PAST PAPERS

5 / 50

PRACTICE QUESTIONS: 3

 Your notes

Came up in: Jan 2020 P2H Q21(b) · Nov 2021 P2H Q21(a)(b) · May 2023 P1HR Q20(b) · Nov 2023 P1H Q24(a)

? What is it?

In plain words

Translation by $\begin{pmatrix} a \\ b \end{pmatrix}$: right a , up b . In the equation: every x becomes $(x - a)$ (opposite!) and $+b$ goes on the end.
Reading it backwards: $5(x + 4)^2 + \dots$ means $a = -4$.

$$y = x^2 + 7x + 20, \begin{pmatrix} 2 \\ 0 \end{pmatrix}$$

$$y = (x - 2)^2 + 7(x - 2) + 20$$

$$y = x^2 + 3x + 10$$

IT LOOKS LIKE

$$\begin{pmatrix} 2 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} 4 \\ 6 \end{pmatrix}$$

$$5(x + 4)^2 + 9(x + 4) + 8$$

EVERY x BECOMES $(x - 2)$

How do I do it?

1 Vector

 $\begin{pmatrix} a \\ b \end{pmatrix}$: right a , up b .

2 Replace

 Every $x \rightarrow (x - a)$, in brackets.

3 Add

 $+b$ at the end.

4 Expand

 Only if the form $ax^2 + bx + c$ is asked.

EVERY $x \rightarrow (x - a)$ Brackets on every x .

V4 Transform the Equation

VARIANT 4 · SEE IT DONE, THEN TRY

 Your notes

WHEN YOU SEE → YOU DO

“Curve **L** is transformed to **S** under the translation $\begin{pmatrix} 2 \\ 0 \end{pmatrix}$ ” → $x \rightarrow (x - 2)$

“You do not need to simplify” → **Stop after substituting**

“ $g(x) = 5(x + 4)^2 + 9(x + 4) + 8$... write down a and b ” → $a = -4$; $b = 8 - 2 = 6$

WORKED EXAMPLE PAST PAPER

May 2023 · Paper 1HR · Q20(b) · 3 marks

Curve **L** has equation $y = x^2 + 7x + 20$. **L** is transformed to **S** under the translation $\begin{pmatrix} 2 \\ 0 \end{pmatrix}$. Find an equation for **S** in the form $y = ax^2 + bx + c$.

STEP 1 $\begin{pmatrix} 2 \\ 0 \end{pmatrix}$: $x \rightarrow (x - 2)$ ◀ right 2

STEP 2 $y = (x - 2)^2 + 7(x - 2) + 20$ ◀ replace every x

STEP 3 $= x^2 - 4x + 4 + 7x - 14 + 20$ ◀ expand

STEP 4 $y = x^2 + 3x + 10$ ◀ collect

ANSWER $y = x^2 + 3x + 10$

✓ $x = 2$ on **S** = $x = 0$ on **L**: 20 ✓

WATCH OUT

✗ $(x + 2)^2 + 7(x + 2) + 20$

✓ $(x - 2)$

Right 2 means $x - 2$ inside.

EXAM TIP

Check a point: **L** at $x = 0$ must equal **S** at $x = 2$.

Check yourself

calculator first · answers on page 15

4.1 Jan 2020 P2H

$y = x^2 + 3x + 4$ is translated by $\begin{pmatrix} 4 \\ 6 \end{pmatrix}$. Find an equation of the new curve.

Answer:

4.2 Nov 2023 P1H

$f(x) = 5x^2 + 9x + 2$ becomes $g(x) = 5(x+4)^2 + 9(x+4) + 8$ by $\begin{pmatrix} a \\ b \end{pmatrix}$. Find a and b .

Answer:

4.3 Nov 2021 P2H

$y = 9 - 3(x + 2)^2$ is translated by $\begin{pmatrix} 4 \\ 0 \end{pmatrix}$. Find the new equation.

Answer:

V5

VARIANT 5

Trig Graphs: Find a , b , c

a = height from the middle line, c = middle line, b = how far the peak moved (opposite sign).

SEEN IN PAST PAPERS

4 / 50

PRACTICE QUESTIONS: 9

 Your notes

Came up in: Jan 2020 P1HR Q22(a) • Jan 2021 P1H Q21(b) • Nov 2021 P2H Q21(d) • Nov 2023 P1H Q24(b)

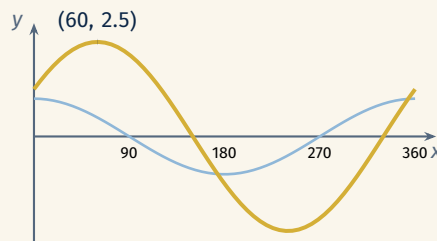
? What is it?**In plain words**

$y = a \cos(x + b) + c$ is $\cos x$ **stretched** by a , **slid** by b (inside: opposite) and **lifted** by c . $\cos x$ peaks at $x = 0$. If the new peak is at $x = 60$, the curve moved right 60, so $b = -60$.

IT LOOKS LIKE

$a \cos(x + b)$

$p \cos(x + q)$

**How do I do it?****1** c

Middle line = $\frac{\max + \min}{2}$.

2 a

$\frac{\max - \min}{2}$ (sign from the shape).

3 b Where is the peak? Moved right $k \rightarrow b = -k$.**4** CheckPut $x = 0$ in your equation; compare with the graph.**MIDDLE, HEIGHT, SLIDE** Inside: opposite sign.

V5 Trig Graphs: Find a, b, c

VARIANT 5 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

" $y = a \cos(x + b)$ " is drawn. Find a and b "

→ **Height and where the peak moved**

"Given that $a > 0$ and $0 < b < 360$ "

→ **Pick the b in that range**

Peak at $x = -45$ for a $\cos$ graph

→ **Moved left 45: $b = +45$** **WORKED EXAMPLE** **PAST PAPER**

January 2020 · Paper 1HR · Q22(a) · 2 marks

The graph of $y = a \cos(x + b)$ for $0 \leq x \leq 360$ has a maximum of 2.5 at $x = 60$ and a minimum of -2.5 at $x = 240$. Find a and b .

STEP 1 $\max = 2.5, \min = -2.5 \Rightarrow a = 2.5$ ◀ height

STEP 2 $\cos$ peaks where $x + b = 0$ ◀ peak rule

STEP 3 $60 + b = 0$ ◀ peak at $x = 60$

STEP 4 $b = -60$ ◀ moved right 60

ANSWER $a = 2.5, b = -60$ ✓ $2.5 \cos(240 - 60) = -2.5$ ✓**WATCH OUT**✗ peak at 60, so $b = 60$ ✓ $b = -60$

Inside the bracket: the opposite sign.

EXAM TIPUse a second feature (the minimum) to check b .**Check yourself**

calculator first · answers on page 15

5.1

Jan 2021 P1H

$y = a \cos(x + b)$, $a > 0$,
 $0 < b < 360$: minimum
 -2 at $x = 90$, crosses up
at $x = 180$. Find a, b .

Answer:

5.2

Nov 2023 P1H

$y = p \cos(x + q)$, $p > 0$,
 $0 < q < 360$: maximum 3
at $x = 315$, minimum -3
at $x = 135$. Find p, q .

Answer:

5.3

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Write down the
maximum point of
 $y = 3 \cos x - 1$ for
 $0 \leq x < 360$.

Answer:

CHECK YOURSELF

Answers



Every answer below was checked with sympy (check_answers.py).

1.1 $(-2, -4); (5, -10)$

1.2 $(-4, 7); (5, 10)$

1.3 $a = 3, b = -11$

2.1 $(-1.5, 4)$

2.2 $(-6, 2)$

2.3 $(-9, 5)$

3.1 $(-3, 5)$

3.2 reflection in the x -axis

3.3 $(-2, -3)$

4.1
 $y = (x - 4)^2 + 3(x - 4) + 10$

4.2 $a = -4, b = 6$

4.3 $y = 9 - 3(x - 2)^2$

5.1 $a = 2, b = 90$

5.2 $p = 3, q = 45$

5.3 $(0, 2)$

THE WHOLE TOPIC ON ONE PAGE

Transformations cheat sheet



Your notes
V1 Translate 21/50

$$f(x + 5) : x - 5$$

Inside opposite; outside as said.

V2 Stretch 16/50

$$f(3x) : x \div 3$$

Inside $\div$; outside $\times$.**V3 Reflect 4/50**

$$-f(x), f(-x)$$

Outside: x-axis. Inside: y-axis.

V4 Equation 5/50

$$x \rightarrow (x - a), +b$$

Every x in brackets.

V5 Trig 4/50

$$a \cos(x + b) + c$$

Middle, height, slide.

**TOP 3 LOST MARKS**

1. $f(x + 5)$ moved right $\times$
2. $f(2x)$ with $x \times 2 \times$
3. $-f(x)$ called a y-axis reflection $\times$

BOTTOM LINE Follow one point · inside: x, opposite · outside: y, as said · minus outside flips in the x-axis.